

CircuitSteer: Geometrically Aligned Multi-Layer Steering via Sparse Autoencoder Circuits

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Actionable Interpretability
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Paper



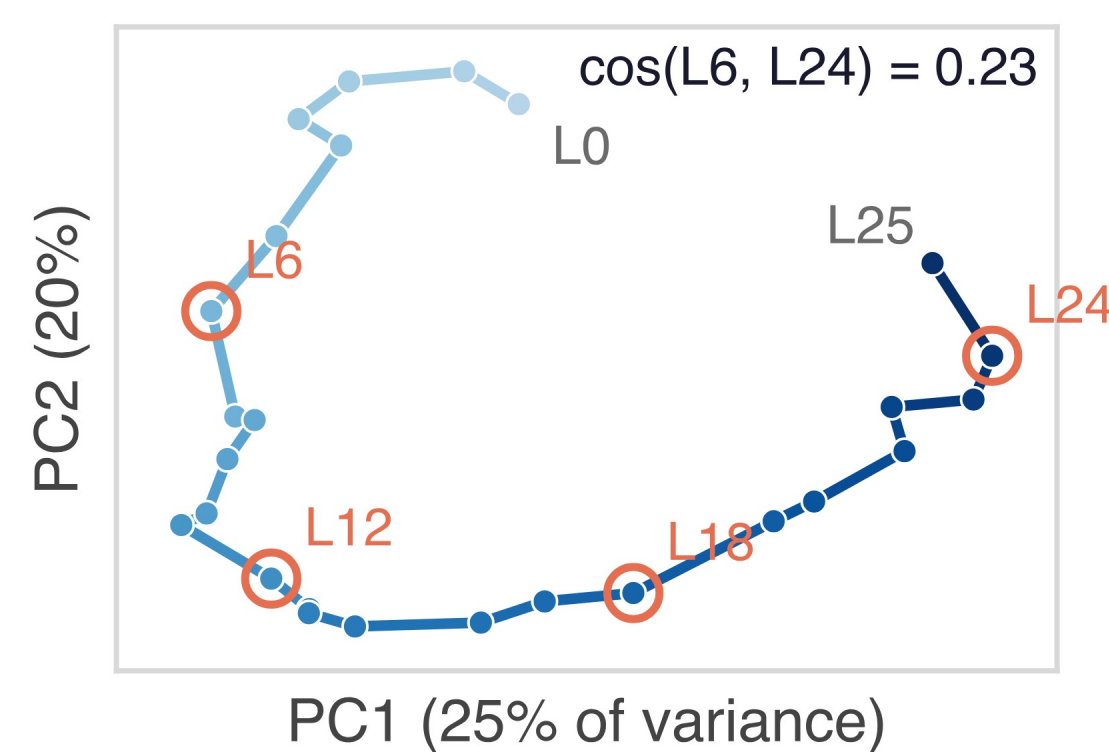
Code

Introduction

A behavior’s direction drifts across layers; in this case, a single-layer push fades, and naive multi-layer steering can cancel itself out. **CircuitSteer** steers a circuit of SAE features that work well together and point the same way across layers. It is the only one of ten methods that significantly reduces the target behavior within the fluency window in all 8 model dataset settings.

MOTIVATION

Steering usually adds one fixed vector at one layer, but a behavior’s direction changes with depth. Here: Gemma-2-2B’s toxicity direction at each of its 26 layers.



0.23 cosine similarity between the toxicity directions at layers 6 and 24

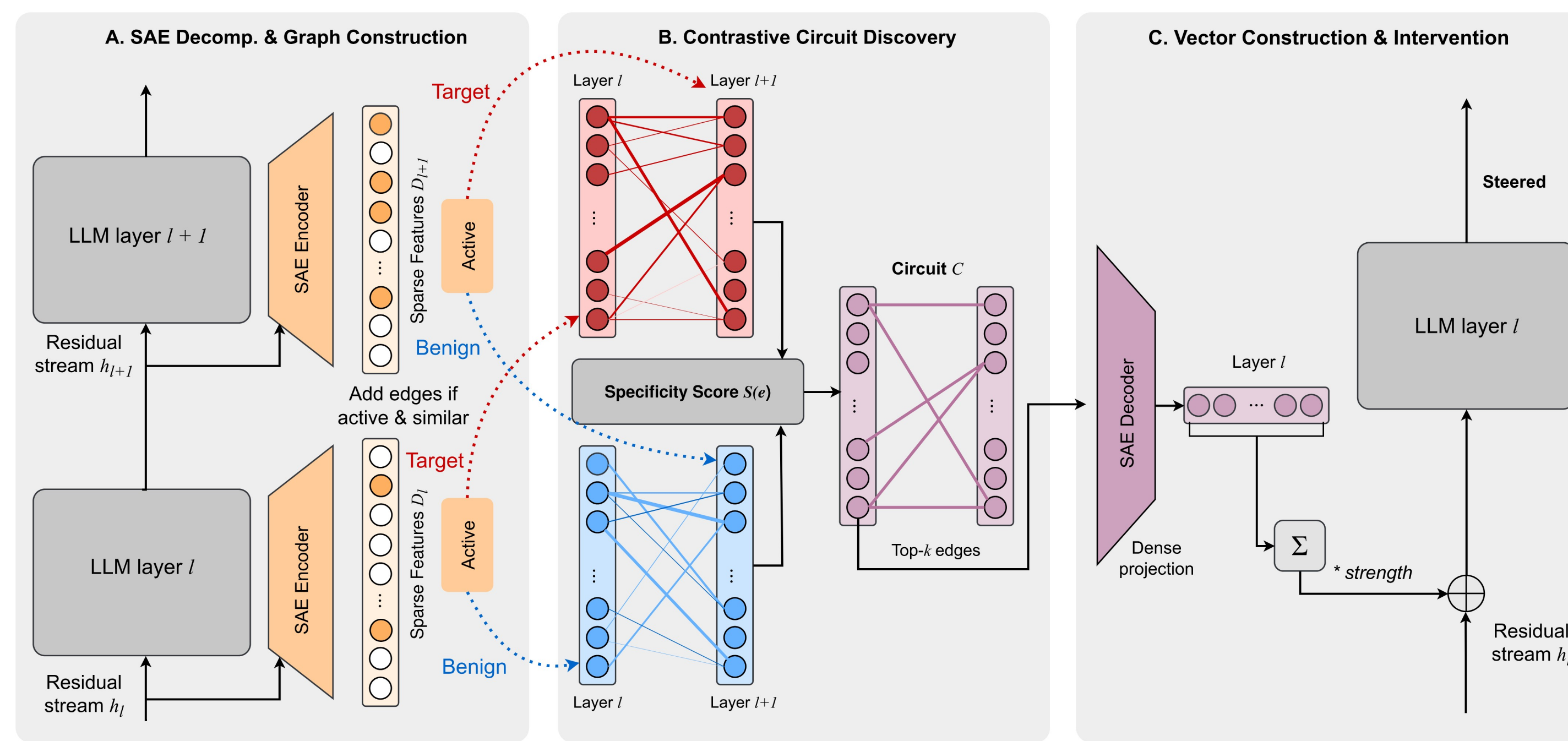
Adjacent layers agree; distant ones point elsewhere.

One dot per layer, projected on two principal components (45% of variance). Orange rings mark the circuit’s SAE layers.

Why naive multi-layer steering breaks

- Misaligned directions.** Vectors built per layer push in partly different directions, so they can cancel rather than add up.
- Compounding perturbations.** Each push changes what every later layer sees, so extra intervention points compound into over-steering and fluency collapse.

METHODOLOGY



A: Link co-active, aligned features

Join a feature at one SAE layer to one at the next when both fire on a prompt ($z > \tau_{\text{act}}$) and their decoder directions agree:

$$\cos(\mathbf{d}_{l,i}, \mathbf{d}_{l',j}) > \tau_{\text{sim}}$$

B: Keep behavior-specific edges

Count each edge on target prompts D_+ and contrast prompts D_- ; keep the top-K edges with $S(e) \geq \tau_{\text{diff}}$:

$$S(e) = P(e | D_+) - P(e | D_-)$$

C: Steer every source layer at once

Sum each layer’s source-feature decoder directions into one dense vector; add every layer’s vector with one shared coefficient λ :

$$\mathbf{v}_l = \sum_{i \in F_l} \mathbf{d}_{l,i} \quad \mathbf{h}'_l = \mathbf{h}_l + \lambda \mathbf{v}_l$$

Training-free: Gemma Scope and Llama Scope SAEs, forward passes only; thresholds fixed in advance, only λ tuned on validation prompts.

RESULTS

Reliable in all settings

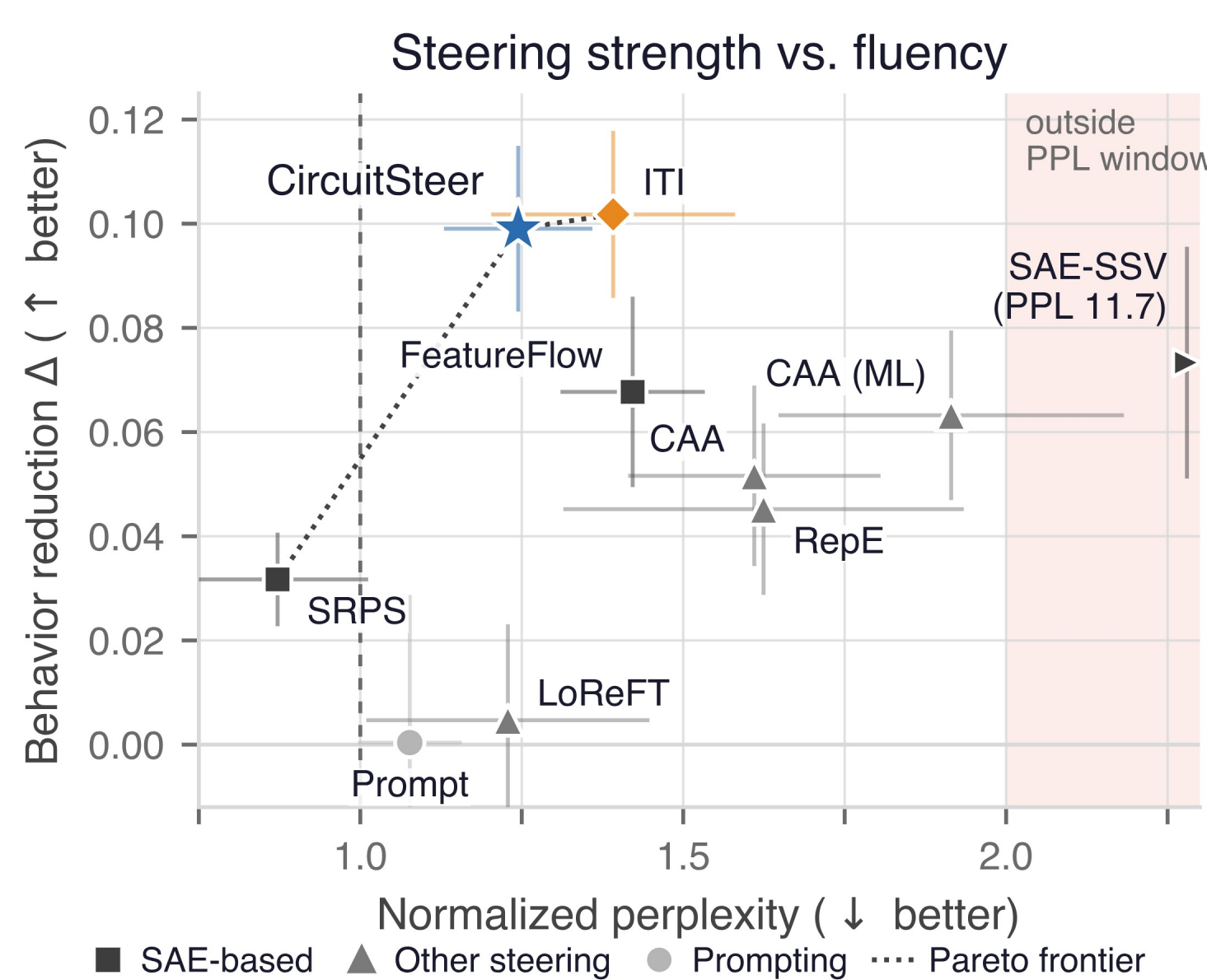
Gemma-2-2B and Llama-3.1-8B-Instruct × toxicity (RealToxicityPrompts, Jigsaw), anger (Emotion) and sycophancy. Fluency window: perplexity ratio ≤ 2.0 vs. the unsteered model; beyond it, an LLM judge rates only 9% of outputs as coherent as unsteered text.

Method	Gemma-2-2B				Llama-3.1-8B				Total
	RTP	Jig.	Emo.	Syc.	RTP	Jig.	Emo.	Syc.	
CircuitSteer	✓	✓	✓	✓	✓	✓	✓	✓	8
ITI	✓	✓	✓	✗	✓	✓	✓	✗	6
CAA (single-layer)	✓	✓	✓	○	✗	✓	○	✓	5
CAA (multi-layer)	✓	✓	✓	○	✓	✓	✗	✗	5
SAE-SSV	✓	✓	✓	–	✗	✗	✗	–	3
FeatureFlow	✓	○	○	○	✓	✗	○	✓	3
Prompting	○	○	○	✓	○	○	○	✓	2
SRPS	○	✓	○	–	○	✓	○	–	2
RepE	✓	○	○	○	✗	○	○	✗	1
LoReFT	○	✗	✗	○	○	○	○	○	0

✓ significant Δ , fluent ○ fluent, not significant ✗ perplexity ratio > 2.0
– no measurable effect

Per-prompt Wilcoxon tests ($n = 75$, Holm-corrected) at each method’s validation-selected λ . Against the strongest baseline per setting, peak Δ is higher or tied in 6 of 8 and lower in 2: the win is coverage.

On the Pareto frontier

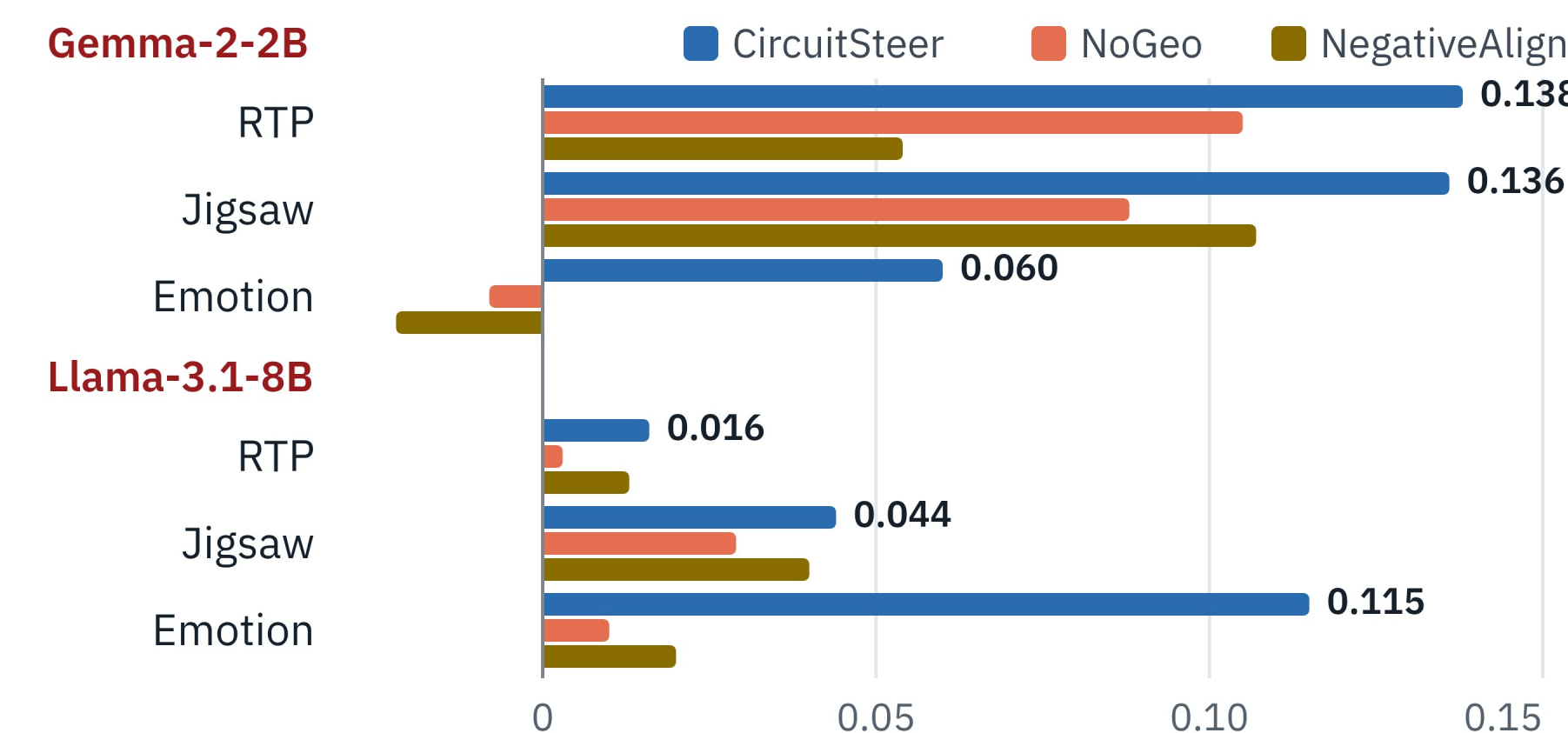


Mean Δ vs. mean perplexity ratio over the 8 settings. CircuitSteer lies on the Pareto frontier, matching the strongest baseline’s Δ at lower perplexity.

- Sycophancy.** On Gemma, $\Delta = 0.134$: nearly twice the best in-window baseline (RepE, 0.071).
- Stable across λ .** On Gemma the perplexity ratio stays 0.91–1.43 up to $|\lambda| = 8$, while multi-layer CAA leaves the window by $|\lambda| = 2$ and RepE by $|\lambda| = 4$. Llama stays fluent only at small $|\lambda|$.
- Scales up.** Qwen3.5-27B with TopK SAEs, thresholds unchanged: perplexity ratio 0.90–1.01, ahead of multi-layer CAA at 3 of 4 λ .

Alignment makes the layers cooperate

Same layers, prompts, λ (–4 Gemma, –3 Llama), and edge count; only the edge rule changes. Removing the alignment test (NoGeo) or reversing it (NegativeAlign) lowers the behavior reduction Δ in every setting, while text stays fluent.



Angle alone: opposing the layer vectors (features, lengths, λ fixed) drives RTP Δ below zero: Gemma 0.082 to –0.030, Llama 0.042 to –0.132.

Knowledge holds; reasoning pays a tax

Steering with	MMLU	GSM8K
Gemma RTP circuit	–1.0 n.s.	–5.4
Gemma Emotion circuit	–0.6 n.s.	–2.6 n.s.
Llama RTP circuit	–4.4 n.s.	–8.8
Llama Emotion circuit	–1.0 n.s.	–6.0

Accuracy change in points, 500 items each; base MMLU 44.2 / 68.8, GSM8K 27.4 / 83.0 (Gemma / Llama). n.s.: $p \geq 0.05$.